

Problem 1

Let $a_n \geq 0$, $n = 1, 2, 3, \dots$, be a sequence such that we have, for some $0 < \gamma < 1$,

$$\limsup_{n \rightarrow \infty} (a_{n+1} - \gamma a_n) \leq 0.$$

Show that $a_n \rightarrow 0$ as $n \rightarrow \infty$.

Solution.

$$S = \limsup_{n \rightarrow \infty} (a_{n+1} - \gamma a_n) = \lim_{n \rightarrow \infty} \sup_{k \geq n} (a_{k+1} - \gamma a_k) \leq 0.$$

Thus $\forall \varepsilon > 0$, there exists a $N \in \mathbb{N}$ such that $\forall n \geq N$ we have that

$$|\sup (a_{k+1} - \gamma a_k : k \geq n) - S| < \frac{\varepsilon}{2}(1 - \gamma)$$

$$S - \varepsilon(1 - \gamma) < \sup (a_{k+1} - \gamma a_k : k \geq n) < S + \frac{\varepsilon}{2}(1 - \gamma) \leq \frac{\varepsilon}{2}(1 - \gamma)$$

Thus $\forall k \geq N$ we have that

$$a_{k+1} - \gamma a_k \leq S + \frac{\varepsilon}{2}(1 - \gamma)$$

$$a_{k+1} \leq S + \gamma a_k + \varepsilon(1 - \gamma) \leq \gamma a_k + \frac{\varepsilon}{2}(1 - \gamma)$$

$$\begin{aligned} a_{N+k} &\leq \gamma a_{N+k-1} + \frac{\varepsilon}{2}(1 - \gamma) \\ &\leq \gamma \left(\gamma a_{N+k-2} + \frac{\varepsilon}{2}(1 - \gamma) \right) + \frac{\varepsilon}{2}(1 - \gamma) \\ &\leq \gamma^k a_N + \frac{\varepsilon}{2}(1 - \gamma) \left(\sum_{m=0}^{k-1} \gamma^m \right) \\ &\leq \gamma^k a_N + \frac{\varepsilon}{2} \end{aligned}$$

Let $K \in \mathbb{N}$ such that $\gamma^K a_N < \frac{\varepsilon}{2}$. Thus $\forall m \geq K + N$,

$$0 \leq a_m \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

.

Thus $a_n \rightarrow 0$.

Problem 2

Let $I \subseteq \mathbb{R}$ be an open interval. We say that a function $f : I \rightarrow \mathbb{R}$ is *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad \text{for all } x, y \in I \text{ and all } \lambda \in [0, 1].$$

(a) Show that for each $x \in I$, the one-sided derivatives of f ,

$$f'_r(x) = \lim_{y \rightarrow x^+} \frac{f(y) - f(x)}{y - x}, \quad f'_\ell(x) = \lim_{y \rightarrow x^-} \frac{f(y) - f(x)}{y - x}$$

exist and are finite.

(b) Show that if $x_1 < x_2$ are points in I , we have

$$f'_\ell(x_1) \leq f'_r(x_1) \leq \frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq f'_\ell(x_2) \leq f'_r(x_2).$$

(c) (Cf.

with Basic Exam, Fall 2014) Let $f_n : I \rightarrow [0, 1]$ be a sequence of convex functions. Show that for each compact interval $J \subseteq I$ there exists a subsequence converging uniformly on J .

Hint. Show first that for all $x < y < z$, $x, y, z \in I$,

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}.$$

Conclude that for every $x \in I$, the difference quotient $(f(x + h) - f(x))/h$ is an increasing function of h when $x + h \in I$ and $h \neq 0$.

Solution. (a) Let $I = (a, b)$. $\forall x, y, z \in I$ where $x < y < z$. $y = \lambda_0 x + (1 - \lambda_0)z = \lambda_1 z + (1 - \lambda_1)x$.

$$\frac{f(z) - f(x)}{z - x} = \frac{\lambda_0 f(z) - \lambda_0 f(x)}{\lambda_0(z - x)}$$

Note the fact that by convexity

$$\begin{aligned} f(y) &\leq \lambda_0 f(x) - \lambda_0 f(z) + f(z) \Rightarrow \lambda_0 f(z) - \lambda_0 f(x) \leq f(z) - f(y) \\ y &= \lambda_0 x + (1 - \lambda_0)z \Rightarrow \lambda_0 z - \lambda_0 x = z - y. \end{aligned}$$

Thus

$$\frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}.$$

Note by convexity:

$$\begin{aligned} f(y) &\leq \lambda_1 f(z) - \lambda_1 f(x) + f(x) \Rightarrow f(y) - f(x) \leq \lambda_1 f(z) - \lambda_1 f(x) \\ y &= \lambda_1 z - \lambda_1 x + x \Rightarrow y - x = \lambda_1 z - \lambda_1 x \end{aligned}$$

Thus

$$\frac{f(z) - f(x)}{z - x} = \frac{\lambda_1(f(z) - f(x))}{\lambda_1(z - x)} \geq \frac{f(y) - f(x)}{y - x}.$$

Thus note that

$$g_x(h) = \frac{f(x+h) - f(x)}{h}$$

is a increasing function for $h \in (a-x, b-x) \cap (\mathbb{R}/\{0\})$. Thus for $0 < h_1 \leq h_2$ we have that $g_x(h_1) \leq g_x(h_2)$, similarly for $h_2 \leq h_1 < 0$ we have that $g_x(h_2) \leq g_x(h_1)$. Thus by Monotone Convergence Theorem:

$$\begin{aligned} f_r'(x) &= \lim_{h \rightarrow 0^+} g_x(h) \\ f_l'(x) &= \lim_{h \rightarrow 0^-} g_x(h) \end{aligned}$$

exists.

(b): Note that we know that $g_x(h)$ is a montonic increasing function when $h \neq 0$, thus for $h > 0$

$$g_x(-h) < g_x(h) \Rightarrow f_l'(x) = \lim_{h \rightarrow 0^+} g_x(-h) \leq \lim_{h \rightarrow 0^+} g_x(h) = f_r'(x)$$

Let $y \in I$ such that $y > x$ and $h_0 = y - x$. We know that since g_x increasing in h ,

$$f_r'(x) \leq g_x(h_0).$$

Furthermore since g_y increasing in h , we know that

$$f_l(y) \leq g_y(-h_0).$$

Thus we have the set of inequalities in the statement.

(c): J is a compact interval so $J = [a, b]$ where $c < a \leq b < d$ where $I = (c, d)$. $\exists c' \in (c, a)$ and $d' \in (b, d)$. Then we know that $\forall x, y \in J$ where $y > x$:

$$\frac{f_n(x) - f_n(c')}{x - c'} \leq \frac{f_n(y) - f_n(x)}{y - x} \leq \frac{f_n(y) - f_n(c')}{y - c'}$$

Furthermore note that

$$\frac{f_n(y) - f_n(c')}{y - c'} \leq \frac{f_n(d') - f_n(y)}{d' - y}.$$

Thus

$$\frac{-1}{a - c'} = \frac{-1}{x - c'} \leq \frac{f_n(y) - f_n(x)}{y - x} \leq \frac{1}{d' - y} = \frac{1}{d - b}.$$

Let

$$L = \max\left(\left|\frac{-1}{a - c'}\right|, \left|\frac{1}{d - b}\right|\right).$$

Thus

$$|f_n(y) - f(x)| \leq L|y - x|.$$

Thus f_n is continuous and all have the same Lipschitz constant. Furthermore, because they are Lipschitz, they are equicontinuous. $\forall \varepsilon > 0$, let $\delta = \frac{\varepsilon}{2L}$ thus

$$|f(y) - f(x)| \leq L|y - x| < \frac{\varepsilon}{2} < \varepsilon$$

for $|y - x| < \delta$. By Arzela Ascoli, there exists a subsequence of f_n which converges uniformly on J .

Problem 3

(Basic Exam, Spring 2018). Consider a sequence $(x_n)_{n=1}^{\infty}$ defined recursively by

$$x_{n+1} = \sin(x_n), \quad n = 1, 2, \dots, \quad x_1 = 1.$$

Show that $\lim_{n \rightarrow \infty} \sqrt{n}x_n$ exists and compute its value.

Hint. Show that $\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2}$ converges to a constant.

Lemma 0.1 (Average of sequence). Let $a_n \geq 0$ sequence where $a_n \rightarrow L$ then

$$\frac{\sum_{k=1}^n a_k}{n} \xrightarrow{n \rightarrow \infty} L$$

Proof: $\forall \varepsilon > 0$, there exists a $N_0 \in \mathbb{N}$ such that $\forall n \geq N_0$ we have that

$$|a_n - L| < \frac{\varepsilon}{2}.$$

Furthermore note that $\exists N_1 \in \mathbb{N}$ such that $\forall n \geq N_1$ there $\frac{1}{n} < \frac{\varepsilon}{2 \sum_{k=1}^{N_0-1} |a_k - L|}$. For all $n \geq \max(N_0, N_1)$ we have the following:

$$\begin{aligned} \left| \frac{1}{n} \sum_{k=1}^n a_k - L \right| &= \frac{1}{n} \left| \sum_{k=1}^n (a_k - L) \right| \\ &\leq \frac{1}{n} \sum_{k=1}^{N_0-1} |a_k - L| + \frac{1}{n} \sum_{k=N_0}^n |a_k - L| \\ &< \frac{\varepsilon}{2} + \frac{n - N_0}{n} \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Thus

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n a_k}{k} = L$$

□

Solution. $x_1 = 1$. Assume that $x_n \in [0, 1]$. Then $x_{n+1} \in [0, 1]$ because $\sin([0, 1]) \subset [0, 1]$. Furthermore $\sin(x)$ is concave down so $x \geq \sin(x)$ hence $x_n \geq x_{n+1}$. Thus x_n is a monotonic

sequence bounded below by 0, thus the limit exists. Let $L = \lim_{n \rightarrow \infty} x_n$. Then since $\sin$ is continuous, $\sin(L) = \lim_{n \rightarrow \infty} \sin(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = L \Rightarrow L = 0$ since $L \in [0, 1]$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2} \right) &= \lim_{n \rightarrow \infty} \frac{x_n^2 - x_{n+1}^2}{x_n^2 x_{n+1}^2} \\ &= \lim_{n \rightarrow \infty} \frac{x_n^2 - \sin^2(x_n)}{x_n^2 \sin^2(x_n)} \\ &= \lim_{n \rightarrow \infty} \frac{x_n^2 - (x_n - \frac{1}{6}x_n^3 + O(x_n^5))^2}{x_n^2 (x_n^2 - O(x_n^3))^2} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{1}{3}x^4 + O(x^6)}{x_n^4 + O(x^6)} = \frac{1}{3} \end{aligned}$$

Then by the lemma above we know that

$$\begin{aligned} \frac{1}{3} &= \lim_{n \rightarrow \infty} \left(\frac{1}{n} \sum_{k=1}^n \left(\frac{1}{x_{k+1}^2} - \frac{1}{x_k^2} \right) \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{1}{nx_{n+1}^2} - \frac{1}{n} \right) \Rightarrow \frac{1}{3} = \lim_{n \rightarrow \infty} \left(\frac{1}{nx_{n+1}^2} \right) = \lim_{n \rightarrow \infty} \left(\frac{1}{nx_{n+1}^2} + \frac{1}{n} \right) - \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right). \end{aligned}$$

Furthermore note that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 &\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{(n+1)x_{n+1}^2} = \frac{1}{3} = \lim_{n \rightarrow \infty} \frac{1}{nx_n^2} \\ 3 &= \frac{1}{\lim_{n \rightarrow \infty} (nx_n^2)^{-1}} \Rightarrow 3 = \lim_{n \rightarrow \infty} \left(((nx_n)^{-1})^{-1} \right) = \lim_{n \rightarrow \infty} (nx_n^2) \end{aligned}$$

Note that $\sqrt{\cdot}$ is continuous on $[0, \infty)$. Thus

$$\sqrt{3} = \sqrt{\lim_{n \rightarrow \infty} nx_n^2} = \lim_{n \rightarrow \infty} \sqrt{nx_n^2}$$

Problem 4

(Basic Exam, Spring 2014). Assume that

$$[0, 1] = \bigcup_{n=1}^{\infty} I_n,$$

where $I_n = [a_n, b_n] \neq \emptyset$ and $I_n \cap I_m = \emptyset$ whenever $n \neq m$.

(a) Let $E = \{a_n; n \geq 1\} \cup \{b_n; n \geq 1\}$ be the set of endpoints of the intervals above. Prove E is closed.

(b) Prove no such family of intervals (I_n) can exist.

Solution. (a): $\forall x \in [0, 1] - E, \exists! n \in \mathbb{N}$ such that $x \in [a_n, b_n]$. Let $\delta = \frac{1}{2} \min(b_n - x, x - a_n)$. Then $U_x = (x - \delta, x + \delta) \subset [a_n, b_n]$ but $a_n, b_n \notin U_x$. Note the following since $I_n \cap I_m = \emptyset$ when $m \neq n$. U_x is not in any other interval. Thus $U_x \notin E$. Hence $U_x \in [0, 1] - E$. Hence $[0, 1] - E$ is open hence E is closed.

(b): Let $b_n < 1$. $\forall \varepsilon \in (0, 1 - b_n), (b_n, b_n + \varepsilon) \subset [0, 1]$.

1. Case 1: $(b_n, b_n + \varepsilon) \subset [a_m, b_m]$ for $a_m > b_n$. This is a contradiction because that means $b_n \in I_m$ which means $I_m \cap I_n \neq \emptyset$. Thus $(b_n, b_n + \varepsilon)$ is not contained in a single interval.
2. Case 2: There exists $I_{m_1}, \dots, I_{m_k}$ such that $I_{m_i} \cap I_n = \emptyset$ where $(b_n, b_n + \varepsilon) \subset \cup_{i=1}^k I_{m_i} = U$. A similar contradiction to 1 occurs as above as the finite union of closed intervals is a closed interval hence $b_n \in U$ contradicting the pairwise disjoint assumption.
3. Case 3: There exists a infinite collection $\{I_{m_k}\}_{k=1}^{\infty}$ such that $(b_n, b_n + \varepsilon)$ is a subset of $\bigcup_{k=1}^{\infty} I_{m_k}$. Thus

$$b_n = \lim_{k \rightarrow \infty} a_{m_k} = \lim_{k \rightarrow \infty} b_{m_k}$$

Hence b_n is a limit point in E . By a similar argument $a_n > 0$ would be a limit point in E . Note the following 0 must be covered so it must be the endpoint of some interval $[0, b_k]$, thus the interior is completely contained in I_k . Thus 0 and similarly 1 is a isolated point in E . Take $E' = E - \{0, 1\}$ (removal of 2 isolated points from a closed set is still closed). Thus each point in E' is a limit point. Thus E' is a closed set where each element of the set is a limit point of the set. $E' = \{a_n : n \in \mathbb{N}, a_n > 0\} \cup \{b_n : n \in \mathbb{N}, b_n < 1\} \Rightarrow E'$ countable. Since E' closed hence it is complete.

$$E' = \bigcup_{\substack{n \in \mathbb{N} \\ a_n \neq 0}} \{a_n\} \cup \bigcup_{\substack{n \in \mathbb{N} \\ b_n \neq 1}} \{b_n\}.$$

The singletons are nowhere dense. By the Baire Category theorem, a complete metric space cannot be the countable union of nowhere dense sets, hence a contradiction occurs and E' and hence E must be uncountable. But that means $[0, 1]$ cannot be the union of countable pairwise disjoint intervals, and must be the union of uncountable such intervals.

Problem 5

(Basic Exam, Spring 2016). Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is a continuously differentiable function. Show that the limit

$$\lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) - n \int_0^1 f(x) dx \right)$$

exists and compute its value.

Solution. We have the following equalities. Note the fourth equality comes from Taylor's Theorem.

$$\begin{aligned}
\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) - n \int_0^1 f(x) \, dx &= n \left[\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) \frac{1}{n} - \int_0^1 f(x) \, dx \right] \\
&= n \left[\sum_{k=0}^{n-1} \left(f\left(\frac{k}{n}\right) \frac{1}{n} - \int_0^{\frac{1}{n}} f\left(x + \frac{k}{n}\right) \, dx \right) \right] \\
&= n \left[\sum_{k=0}^{n-1} \int_0^{\frac{1}{n}} \left(f\left(\frac{k}{n}\right) - f\left(x + \frac{k}{n}\right) \right) \, dx \right] \\
&= -n \left[\sum_{k=0}^{n-1} \int_0^{\frac{1}{n}} \left(f'\left(\frac{k}{n}\right) x + \int_0^x (x-t) f''(t) \, dt \right) \, dx \right] \\
&= -\frac{1}{2} n \left[\sum_{k=0}^{n-1} \frac{1}{n^2} f'\left(\frac{k}{n}\right) + \int_0^{\frac{1}{n}} \int_0^x (x-t) f''(t) \, dt \, dx \right] \\
&= -\frac{1}{2} \sum_{k=0}^{n-1} \frac{1}{n} f'\left(\frac{k}{n}\right) - \frac{n}{2} \int_0^{\frac{1}{n}} \int_0^x (x-t) f''(t) \, dt \, dx
\end{aligned}$$

Note that

$$\left| \frac{n}{2} \int_0^{\frac{1}{n}} \int_0^x (x-t) f''(t) \, dt \, dx \right| \leq \frac{1}{12n^2} \|f''\|_{\infty}$$

Thus we have:

$$\left| \left(\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) - n \int_0^1 f(x) \, dx \right) - \frac{f(0) - f(1)}{2} \right| \leq \frac{1}{2} \left| \sum_{k=0}^{n-1} \frac{1}{n} f'\left(\frac{k}{n}\right) - \int_0^1 f'(x) \, dx \right| + \frac{1}{12n^2} \|f''\|_{\infty}$$

f is a continuous function on a compact interval and hence is Riemann Integrable on the interval.

Thus $\exists N_0 \in \mathbb{N}$ such that for all $n \geq N_0$

$$\left| \sum_{k=0}^{n-1} \frac{1}{n} f\left(\frac{k}{n}\right) - \int_0^1 f(x) \, dx \right| < \varepsilon.$$

$\exists N_1 \in \mathbb{N}$ such that for all $n \geq N_1$

$$\frac{\|f''\|_{\infty}}{12n^2} < \frac{\varepsilon}{2}$$

Thus $\forall n \geq \max(N_0, N_1)$ we have that

$$\left| \left(\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) - n \int_0^1 f(x) \, dx \right) - \frac{f(0) - f(1)}{2} \right| < \varepsilon.$$

Thus

$$\lim_{n \rightarrow \infty} \left(\sum_{k=0}^{n-1} f\left(\frac{k}{n}\right) - n \int_0^1 f(x) dx \right) = \frac{f(0) - f(1)}{2}$$

Problem 6

(Basic Exam, Spring 2023). Let $\varepsilon > 0$ and let f be a continuous function on $[0, \frac{\pi}{2}]$ such that $f(\frac{\pi}{2}) = 0$ and

$$|f(x)^2 + \sin x f(x)| \leq \varepsilon^2, \quad \text{for all } 0 \leq x \leq \frac{\pi}{2}.$$

Show that for small enough $\varepsilon > 0$,

$$\max_{0 \leq x \leq \frac{\pi}{2}} |f(x)| \leq 3\varepsilon.$$

Solution. For all $x \in [0, \frac{\pi}{2}]$ we have

$$|f(x)| |f(x) + \sin(x)| = |f(x)^2 + \sin(x)f(x)| \leq \varepsilon^2$$

Thus

$$f(x) \in [-\varepsilon, \varepsilon] \cup [-\sin x - \varepsilon, -\sin x + \varepsilon] = I_1 \cup I_2(x).$$

Note that $I_1(x) \cap I_2(x) = \emptyset \Leftrightarrow \sin x > 2\varepsilon$. Thus for $\sin x \gg \varepsilon$, $f(x)$ must fall cleanly in one and not the other. Let $x_0 \in [0, \frac{\pi}{2}]$ be the unique point such that $\sin(x_0) = 2\varepsilon$. Thus $\sin([0, x_0]) \subset [0, 2\varepsilon]$ and then $\sin((x_0, \frac{\pi}{2})) \subset (2\varepsilon, 1]$.

1. $x \in [0, x_0]$: Thus $I_1 \cap I_2(x) \neq \emptyset$. Thus $f(x) \in [-\sin(x) - \varepsilon, \varepsilon] \subseteq [-3\varepsilon, 3\varepsilon]$. Thus $|f(x)| \leq 3\varepsilon$.
2. $x \in (x_0, \frac{\pi}{2}]$. Let $E = \{x \in (x_0, \frac{\pi}{2}] : f(x) \in I_1\}$. $\frac{\pi}{2} \in E$.

1. **Closed:** f is continuous. $E = f^{-1}([-\varepsilon, \varepsilon]) \cap (x_0, \frac{\pi}{2}]$ hence E is closed in $(x_0, \frac{\pi}{2}]$.
2. **Open:** Let $x \in E$. $x > x_0$ thus $d(x) = \sin(x) - 2\varepsilon > 0$ the distance between I_1 and $I_2(x)$. $f(x) > -\varepsilon$ by assumption. $\exists \delta_0 > 0$ such that

$$|\sin(x) - \sin(y)| < \frac{d(x)}{3} \Rightarrow \sin(y) > \sin(x) - \frac{d(x)}{3}.$$

By continuity of f , $\exists \delta_1 > 0$ such that $|f(x) - f(y)| < \frac{2}{3}(d(x))$.

Let $\delta = \min(\delta_1, \delta_2, x - x_0)$. Let $y \in (x - \delta, x + \delta)$

Assume for contradiction $f(y) \in I_2(y)$. Thus

$$f(y) \leq -\sin(y) + \varepsilon.$$

Thus

$$\begin{aligned}
f(x) - f(y) &\geq \sin(y) - 2\varepsilon \\
&> \sin(x) - \frac{d(x)}{3} - 2\varepsilon \\
&= \sin(x) - 2\varepsilon - \frac{1}{3}(\sin(x) - 2x) = \frac{2}{3}(\sin(x) - 2\varepsilon) = \frac{2}{3}d(x) > 0
\end{aligned}$$

But we know that

$$|f(x) - f(y)| < \frac{2}{3}d(x).$$

Thus we have a contradiction, and $f(y) \in I_1$. Hence $(x - \delta, x + \delta) \subseteq E$. Thus E is open.

Thus $E \subseteq (x_0, \frac{\pi}{2}]$ is a nonempty subset which is both open and closed in $(x_0, \frac{\pi}{2}]$ hence by connectedness $E = (x_0, \frac{\pi}{2}]$. Thus for $x \in (0, \frac{\pi}{2}]$ we have that $|f(x)| < \varepsilon < 3\varepsilon$. Thus

$$\max_{0 \leq x \leq \frac{\pi}{3}} f(x) < 3\varepsilon$$